

PUYSEGUR POINT, New Zealand

WMO: 938050

Lat: 46.156S

Lon: 166.613E

Elev: 44

StdP: 100.80

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.4	4.2	-1.4	3.4	5.2	-0.2	3.7	5.5	24.0	10.7	21.3	11.7	4.7	40	0.836

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.3	20.4	16.0	19.2	15.4	18.3	14.9	17.0	19.4	16.2	18.3	15.5	17.7	12.1	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.5	18.1	15.2	10.8	17.4	14.4	10.3	16.9	48.1	19.4	45.6	18.5	43.6	17.6	20.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	19.9	18.0	1.9	22.7	1.3	1.5	1.0	23.8	0.3	24.6	-0.5	25.5	-1.4	26.6
			0.4	18.1	1.1	1.0	-0.4	18.8	-1.1	19.4	-1.7	20.0	-2.5	20.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	13.9	14.1	13.4	12.0	10.7	8.8	8.5	8.8	9.6	10.1	10.8	12.7		
	DBStd	2.88	2.21	2.25	2.19	2.13	2.19	1.87	1.97	1.89	2.01	1.99	2.12	2.28		
	HDD10.0	239	0	0	1	6	18	46	56	47	30	22	13	2		
	HDD18.3	2645	138	119	155	191	238	287	306	296	261	255	225	173		
	CDD10.0	640	121	116	105	64	38	9	8	9	19	26	38	87		
	CDD18.3	3	1	2	1	0	0	0	0	0	0	0	0	0		
	CDH23.3	1	0	1	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	7.8	8.6	7.9	7.9	7.9	8.1	7.3	6.9	6.9	7.8	8.2	8.3	8.0		
(15) Precipitation	PrecAvg	2363	214	173	208	217	214	200	186	166	185	206	194	175		
	PrecMax	2857	416	332	423	427	384	358	277	261	306	335	356	285		
	PrecMin	1595	77	55	83	87	65	93	44	9	63	102	62	28		
	PrecStd	294	77	74	82	93	73	72	55	53	67	66	80	60		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	22.4	20.7	18.8	17.6	15.0	14.6	14.8	16.6	17.6	18.6	20.8		
		MCWB	16.6	17.5	15.9	14.1	13.8	11.8	10.9	10.4	11.3	12.6	13.7	15.8		
	2%	DB	20.0	20.4	19.5	17.4	16.0	13.4	13.3	13.4	15.0	15.9	16.9	18.9		
		MCWB	16.0	16.3	15.8	13.5	12.8	11.0	10.0	9.9	11.1	12.0	13.2	14.8		
	5%	DB	18.8	19.0	18.2	16.5	14.9	12.5	12.4	12.4	13.8	14.6	15.6	17.8		
		MCWB	15.4	15.6	15.2	13.3	12.3	10.6	9.7	9.7	10.4	11.5	12.6	14.4		
	10%	DB	17.7	17.9	17.0	15.5	14.0	11.6	11.6	11.6	12.7	13.3	14.4	16.6		
		MCWB	14.8	15.0	14.3	12.9	11.9	9.9	9.4	9.3	10.0	10.8	11.9	13.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	18.5	17.7	15.5	14.9	12.5	11.7	11.4	12.5	13.5	14.9	16.7		
		MCDB	19.5	21.3	19.5	17.2	16.8	14.1	13.5	12.9	15.0	15.8	17.1	19.4		
	2%	WB	16.8	17.1	16.5	14.5	13.5	11.5	10.9	10.6	11.6	12.7	14.0	15.7		
		MCDB	19.3	19.4	18.3	16.3	15.0	13.0	12.5	12.3	13.8	15.0	15.9	17.9		
	5%	WB	15.9	16.1	15.6	13.8	12.7	10.8	10.2	10.1	10.9	11.9	13.1	14.9		
		MCDB	18.2	18.3	17.6	15.7	14.3	12.2	11.8	11.9	13.2	14.1	15.1	17.3		
	10%	WB	15.2	15.3	14.6	13.1	12.0	10.1	9.6	9.6	10.2	11.1	12.2	14.1		
		MCDB	17.4	17.4	16.8	15.1	13.8	11.4	11.3	11.3	12.4	13.0	14.2	16.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.3	4.2	4.0	3.5	3.6	3.8	3.8	3.9	4.0	4.0	4.3		
		MCDBR	6.0	5.9	5.7	5.2	4.3	4.2	4.4	4.6	5.2	5.6	5.6	6.1		
	5% WB	MCWBR	3.9	3.9	3.8	3.8	3.3	3.5	3.3	3.2	3.6	3.9	3.8	4.0		
		MCWBR	5.1	5.3	4.9	4.6	3.8	3.9	3.7	4.0	4.7	4.8	4.9	5.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.346	0.341	0.329	0.316	0.319	0.311	0.306	0.313	0.342	0.339	0.335	0.336		
		taud	2.538	2.564	2.603	2.624	2.560	2.533	2.574	2.581	2.490	2.514	2.544	2.571		
	Ebn at Noon	Ebn at Noon	973	948	911	845	748	709	756	833	878	939	978	991		
		Edh at Noon	105	97	84	69	60	55	58	70	92	101	104	103		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	5.58	4.74	3.46	2.22	1.29	0.99	1.19	1.94	2.98	4.24	5.29	5.86		
	RadStd	RadStd	0.45	0.32	0.23	0.16	0.09	0.10	0.10	0.11	0.20	0.30	0.43	0.39		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	+0.65	N/A	N/A	N/A	-42	-128	+78	N/A

Nomenclature: See separate page